2. LITERATURE REVIEW

The development of soft robotic grippers has increased the need for sensing systems that can provide reliable information about contact forces, deformation, object properties, and gripping conditions. Unlike conventional rigid grippers, soft grippers are designed using compliant and deformable materials, which allow them to adapt to objects having different shapes and surface characteristics. However, this flexibility also creates challenges in sensing, since the sensors must deform along with the gripper without significantly affecting its motion or gripping performance.

Recent research has therefore focused on developing flexible and highly sensitive tactile and proprioceptive sensing technologies for soft robotic systems. Different sensor technologies have been investigated based on their working principles and their ability to measure static force, dynamic force, strain, slip, surface texture, and object geometry. The literature reviewed in this work mainly focuses on capacitive, piezoresistive, piezoelectric, optical, strain, and ionic sensing technologies and their suitability for soft robotic grippers.

2.1 Advanced Tactile Sensing Technologies for Soft Grippers

The author presented a comprehensive review of advanced tactile sensing technologies used in soft robotic grippers. The authors classified tactile sensors according to their sensing mechanisms into capacitive and piezoresistive, piezoelectric and triboelectric, and optical sensing technologies.

Capacitive and piezoresistive sensors are mainly used for detecting direct pressure and strain through changes in capacitance or electrical resistance. These sensors are particularly useful when the gripper needs to determine the magnitude of a contact force. Piezoelectric and triboelectric sensors, on the other hand, are more suitable for detecting dynamic mechanical events such as vibration, impact, and the initial stages of object slipping. Optical sensing technologies, including Fiber Bragg Grating (FBG) and vision-based tactile sensors such as GelSight, provide information about deformation, surface texture, and multi-directional forces.

The review also emphasizes the development of new sensing materials, including carbon nanotubes, liquid metals, and conductive hydrogels. These materials can be incorporated into flexible structures and are capable of undergoing significant deformation while maintaining their sensing functionality. Such material developments are important for soft grippers because conventional rigid sensors may restrict the flexibility of the actuator.

One of the major challenges identified in the study is the integration of sensors with the soft actuator. The sensing elements need to be embedded into the gripper without reducing its flexibility or interfering with its operation. Another important limitation is hysteresis, where the soft sensing material requires time to return to its original state after being compressed or deformed. This can introduce delays and inaccuracies into the sensor output. The complex

and nonlinear nature of signals generated by soft sensors also creates a need for advanced data-processing and machine-learning techniques.

An important observation from this review is that a single sensor type may not be sufficient to reproduce the sensing capabilities required for a soft gripper. The authors therefore highlight multimodal sensing, where different sensors are combined according to their individual strengths. For example, a piezoresistive sensor can be used for measuring static gripping force while a piezoelectric sensor can be used to detect slip. This combination allows the system to respond both to the magnitude of the applied force and to dynamic changes occurring at the contact surface.

The review identifies three important applications of tactile sensing: force measurement, object property perception, and slip detection. Force measurement requires the detection of both normal and shear forces so that the gripper can apply an appropriate gripping force. Object property perception involves using sensor arrays to obtain information about an object's shape, size, stiffness, and surface texture. Slip detection is particularly important for autonomous gripping, as dynamic sensors can detect the micro-vibrations associated with an object beginning to slip and allow the gripping force to be adjusted accordingly.

2.2 Human-Centered Sensor Technologies for Soft Robotic Grippers

The author conducted a comprehensive review of human-centered sensor technologies for soft robotic grippers. The study used the PRISMA methodology to systematically examine and organize existing research in the field. The objective was to provide a structured understanding of different sensor technologies and their suitability for particular sensing requirements.

The authors classified sensor technologies according to the type of information that needs to be detected. Piezoelectric sensors were identified as suitable for dynamic touch applications such as detecting slippage, impact, and micro-vibrations. Since their response is associated with changes in mechanical stress, they are suitable for situations requiring rapid detection of dynamic events. However, they are not considered suitable for measuring a constant static gripping force because their output decreases when the applied pressure remains unchanged.

Piezoresistive and capacitive sensors were identified as more appropriate for static touch and grip-force measurement. Piezoresistive sensors offer high sensitivity and can be made highly stretchable, making them suitable for flexible gripper structures. Capacitive sensors can provide stable measurements over longer periods, although their performance can be affected by electrical interference from the surrounding environment.

Optical tactile sensors were discussed for applications where detailed spatial information is required. These sensors can use miniature cameras or optical structures to determine the three-dimensional shape and surface characteristics of an object. Their high spatial resolution makes them useful for recognizing object geometry and texture through touch. However, the

associated increase in physical size and computational requirements can be a limitation for compact soft grippers.

The study also identifies important characteristics that sensors should possess for soft robotic applications. Compliance and flexibility are necessary so that the sensor can bend and stretch together with the soft actuator. Multifunctionality is another desirable characteristic, allowing a single sensing system to detect multiple physical quantities such as pressure, strain, and temperature. In addition, the surface and material properties of the sensor should not negatively affect the gripping action or damage the object being handled.

Overall, this study provides a useful basis for selecting sensors according to their intended function. Static force measurement, dynamic slip detection, and shape or texture recognition require different sensing mechanisms. Therefore, sensor selection should be based on the specific requirements of the gripper rather than on the sensitivity of a particular sensor alone.

2.3 Resistive Sensing Technologies in Soft Robotic Grippers

The author reviewed resistive sensing technologies for soft robotic grippers, with particular attention to strain, tactile, and ionic sensors. The study points out that soft robotic grippers require both external and internal sensing information to operate safely and autonomously.

Exteroceptive sensing provides information about the external environment, such as contact with an object, while proprioceptive sensing provides information about the internal state of the gripper, such as the bending or deformation of its fingers. Resistive sensing technologies are particularly relevant because they can be implemented using flexible materials and generally have relatively simple electrical interfaces.

The authors categorize the main resistive sensing technologies into strain sensors, tactile sensors, and ionic sensors. Strain sensors can be used to monitor deformation and the actuation state of a soft gripper, including bending and stretching of its fingers. Tactile sensors are intended to detect physical contact and pressure distribution and can also be used for identifying the onset of object slipping. Ionic sensors are identified as an emerging technology with the potential to detect mechanical and environmental stimuli simultaneously.

The study also discusses the use of hybrid material systems involving polymers, elastomers, and gels. Such combinations can improve mechanical durability and reduce problems associated with material degradation. Additive manufacturing and three-dimensional printing are also highlighted as potential methods for embedding sensors directly into soft robotic structures. This can improve integration and potentially reduce manufacturing cost and variation.

Despite these developments, several limitations continue to restrict the widespread practical use of resistive sensors. Hysteresis is a major issue because the sensing material may not immediately recover its original shape after being stretched or compressed. This affects the accuracy of measurements, particularly during repeated or dynamic operation. Ionic and

hydrogel-based sensors can also suffer from dehydration and leakage when exposed to ambient conditions for long periods, resulting in reduced electrical conductivity and long-term instability.

Another important challenge is signal processing. Since flexible sensing materials undergo deformation and generate nonlinear and sometimes noisy electrical signals, conventional control methods may not always be sufficient. The study therefore identifies deep-learning and artificial-intelligence-based signal interpretation as potential approaches for processing complex sensor signals in real time.

2.4 Comparison and Synthesis of Sensor Technologies

The reviewed studies indicate that different sensor technologies provide different types of information and therefore have different areas of application. Capacitive and piezoresistive sensors are mainly suitable for static contact and force measurement, whereas piezoelectric sensors are more effective for dynamic events such as vibration, impact, and slip. Optical sensors provide detailed information about object geometry and surface texture but generally require more complex hardware and data processing.

Sensor Type	Primary Measurement	Major Advantage	Major Limitation
Capacitive	Touch, force, proximity	Stable sensing and high sensitivity	Can be affected by electrical interference
Piezoresistive	Pressure, strain, static force	Simple interface, high sensitivity and stretchability	Hysteresis and nonlinear response
Piezoelectric	Dynamic force, vibration, slip	Fast response to dynamic events	Not suitable for continuous static-force measurement
Optical	Shape, texture, deformation	High-resolution spatial information	Higher computational and integration requirements
Strain/Bending	Finger deformation and posture	Provides proprioceptive information	Measures deformation rather than direct contact force
Ionic	Mechanical and environmental stimuli	Potential for multimodal sensing	Long-term stability and dehydration issues

A common conclusion across the reviewed literature is that **multimodal sensing is more effective than depending on a single sensor technology**. Static force, dynamic slip, finger deformation, and object geometry represent different sensing requirements. A combination of sensors can therefore provide more complete feedback to the gripper. For example, piezoresistive sensing can be used to determine the applied gripping force, while piezoelectric sensing can provide rapid slip detection. Internal strain or bending sensing can additionally provide information about the position and deformation of the soft fingers.

2.5 Research Gaps and Practical Considerations

Although significant progress has been made in flexible and tactile sensing, the literature identifies several challenges that remain important for practical implementation. Sensor-actuator integration is one of the primary concerns because the sensing elements must maintain flexibility without reducing the performance of the soft gripper. Hysteresis and nonlinear sensor responses can affect measurement accuracy and repeatability. Long-term stability is also a concern for materials such as hydrogels and ionic sensors.

Another challenge is the interpretation of sensor data. Flexible sensors can generate complex signals because the sensing material itself undergoes deformation. As a result, reliable operation may require appropriate signal-processing techniques and, in advanced systems, machine-learning-based interpretation.

The reviewed studies also show that sensor selection should be application-specific. A sensor that performs well for static force measurement may not be suitable for detecting slip, while a sensor capable of detecting dynamic events may not provide reliable continuous force measurements. Therefore, the selection of sensors must consider the required measurement, response characteristics, integration method, flexibility, durability, and cost.

2.6 Relevance to the Present Project

The reviewed literature provides the basis for selecting and integrating sensors for the present soft robotic gripper project. The studies demonstrate that sensing is required not only to determine the contact force between the gripper and the object but also to obtain information about the deformation and operating condition of the gripper.

For the present work, external tactile sensors can be considered for detecting contact and estimating gripping force, while internal sensing can provide information about finger bending or deformation. The literature also suggests that dynamic sensing can be useful for detecting slip and preventing accidental dropping of an object. Thus, combining different sensing mechanisms can provide more useful feedback than relying on a single sensor.

The practical sensor options identified during the literature study include capacitive, piezoresistive, optical, piezoelectric, internal strain/bending, fluid-pressure, and actuator tension sensing. Their approximate costs were also recorded as part of the present study. This

provides a basis for considering not only the technical suitability of each sensor but also its feasibility for implementation in the project.

Overall, the literature indicates that an effective sensing system for a soft robotic gripper should provide adequate force and deformation information while remaining flexible, compact, reliable, and sufficiently simple to integrate with the actuator and data-acquisition system. The findings from the reviewed studies therefore guide the sensor selection and system design adopted in the present project.